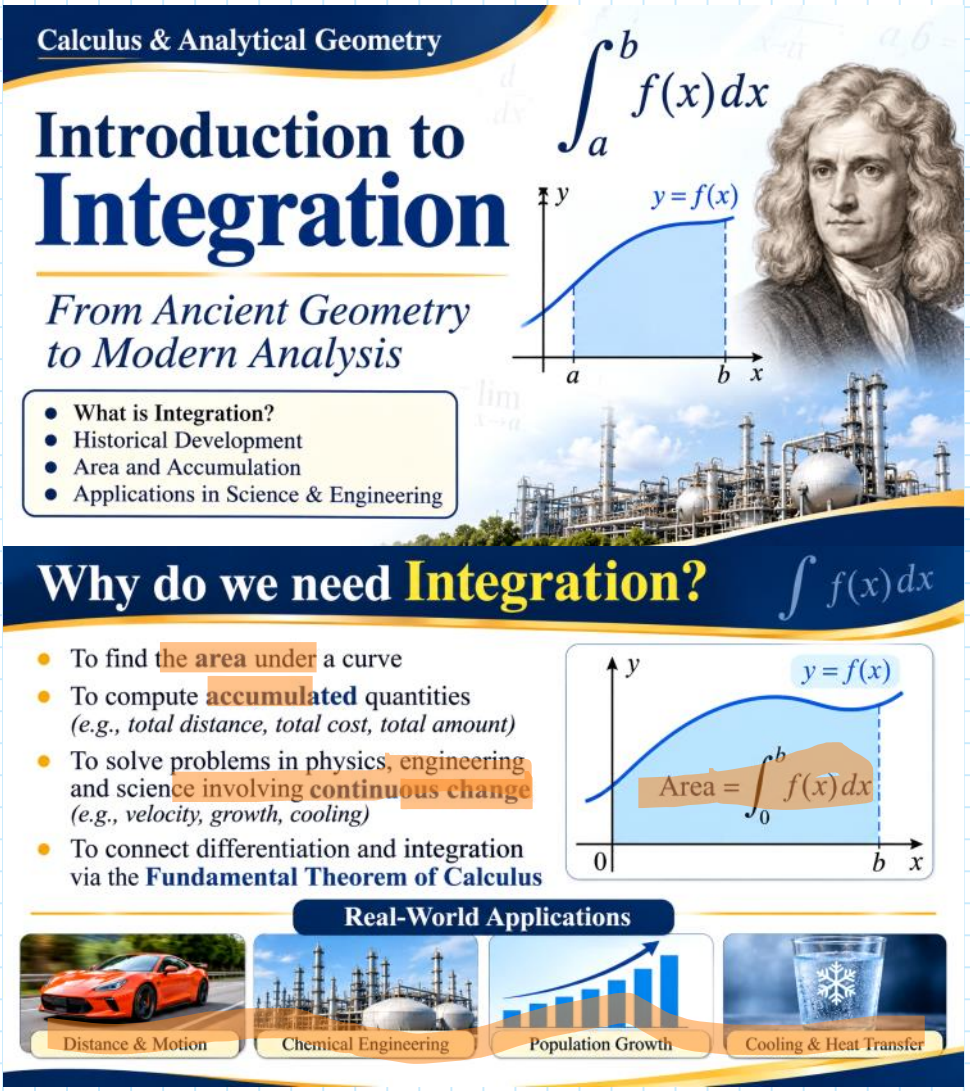
Integration-I(Riemann Sum & Theorems)

Monday, 2 November 2026 9:23 pm





1. Ancient Precursors: The Method of Exhaustion

- **Ancient Egypt & Babylon (c. 1800 BCE):** Early practitioners calculated volumes and areas for practical tasks, such as the Moscow Mathematical Papyrus containing formulas for a pyramidal frustum.
- **Ancient Greece (c. 5th–3rd Century BCE):** Antiphon and **Eudoxus of Cnidus** developed the **method of exhaustion**, inscribing and circumscribing polygons to approximate areas like a circle.
- **Archimedes:** Extended Eudoxus's work to rigorously find the areas and volumes of parabolas, spheres, and spirals using infinitesimal slices and mechanical levers.
- **Asia & the Middle East:** Around the 3rd to 5th centuries, Chinese mathematicians like **Liu Hui**, **Zu Chongzhi**, and **Zu Geng** used polygon limits to find circle areas and sphere volumes. In the 11th century, **Ibn al-Haytham** (Alhazen) worked out sums of fourth powers related to volume problems. Scribd +5

2. The 17th Century: The Birth of Calculus

- **Indivisibles and Infinitesimals:** Bonaventura Cavalieri introduced the *method of indivisibles*, treating areas as sums of lines and volumes as sums of areas. Pierre de Fermat, John Wallis, and Isaac Barrow pushed these concepts further. YouTube · TG Campus IN +4
- **Newton and Leibniz:** **Sir Isaac Newton** and **Gottfried Wilhelm Leibniz** independently formulated the core principles of calculus. They recognized that differentiation and integration are inverse operations via the **Fundamental Theorem of Calculus**. Leibniz introduced the modern integral sign ($\int$) and the term "integral" (coined with help from Johann Bernoulli). Wikipedia +4

3. The 19th Century: Rigor and Limits



Augustin-Louis Cauchy: Put calculus on a formal foundation by defining the definite integral of a continuous function as a limit of sums.



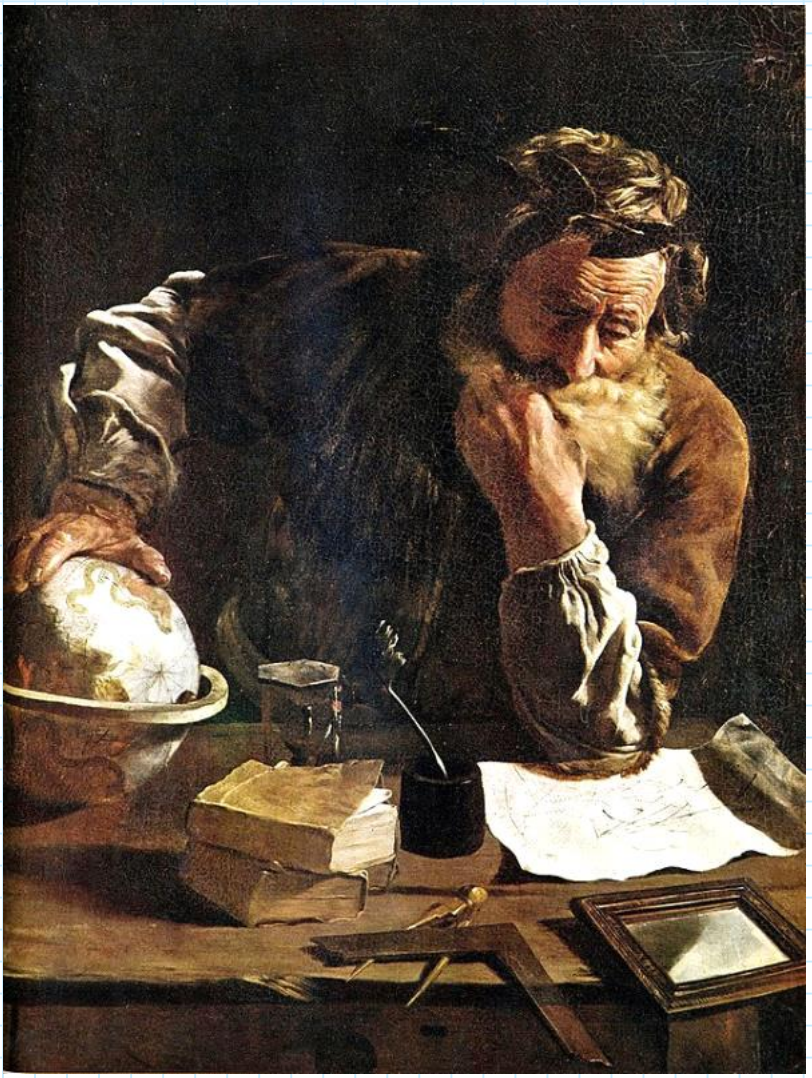
Bernhard Riemann: Generalized Cauchy's definition in 1854 to handle more discontinuous or bounded functions using upper and lower sums (the **Riemann integral**). YouTube · Eddie Woo +4

4. The 20th Century: Modern Measure Theory



Henri Lebesgue: Introduced the **Lebesgue integral** in 1902, which used measure theory to integrate highly erratic or discontinuous functions (such as the Dirichlet function) where Riemann integration failed. Persée +3

Area and Estimating with Finite Sums

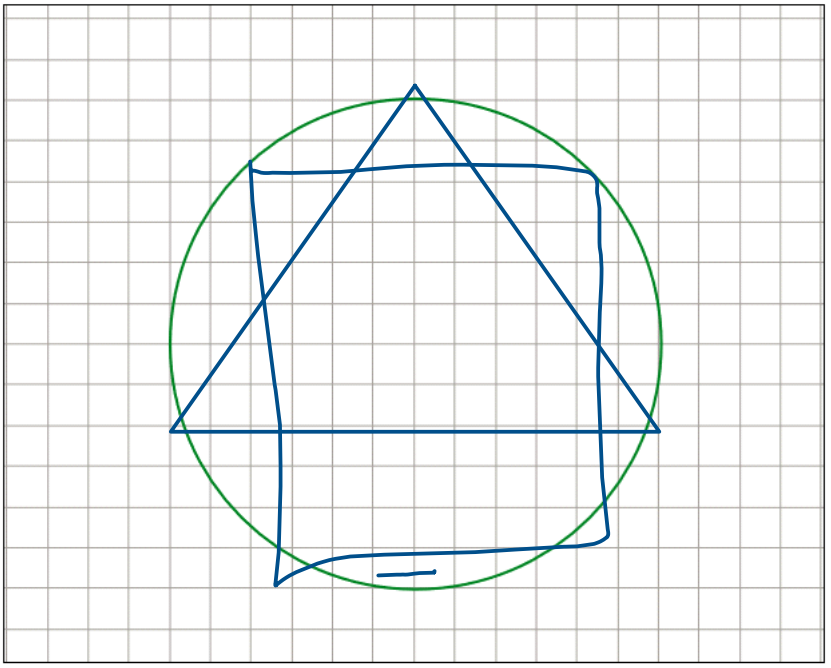


According to the most popular account, Archimedes was contemplating a mathematical diagram when the city was captured. A Roman soldier commanded him to come and meet Marcellus, but he declined, saying that he had to finish working on the problem. This enraged the soldier, who killed Archimedes with his sword.

Area of a Circle:

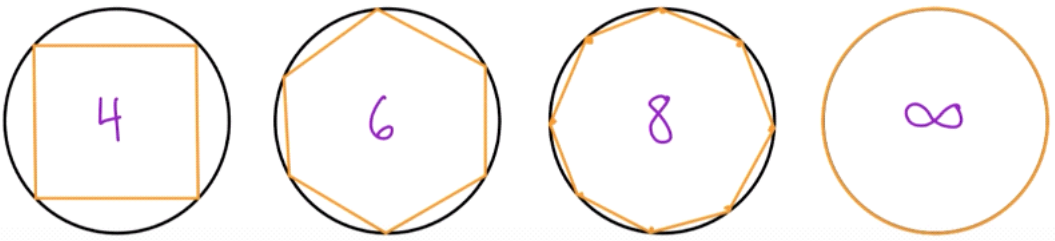
A personal Approach

Find a way to approximate the area of the circle below without using any prior knowledge of a formula for the area. You may use your knowledge of areas of rectangles, triangles and other polygons. Document your process clearly and come up with a good approximation. You can assume one of the squares on the grid to be 1 sq. unit.

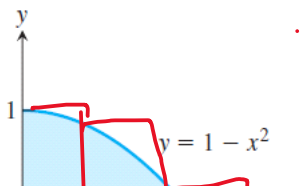


The Method of Exhaustion

...the **Method of Exhaustion** builds on this to find area of curved shapes:



- Integral calculus is directly based on this method.
- Calculus in general relies on the idea of approaching infinity.
  - Leibniz gave field the name "calculus of the infinitesimals"



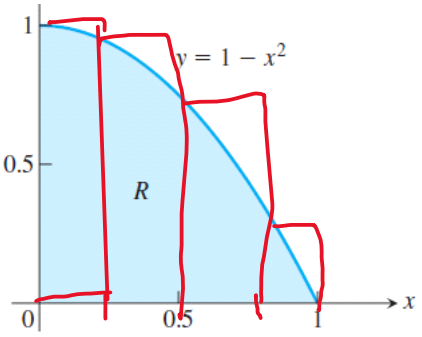


FIGURE 5.1 The area of the region R cannot be found by a simple formula.

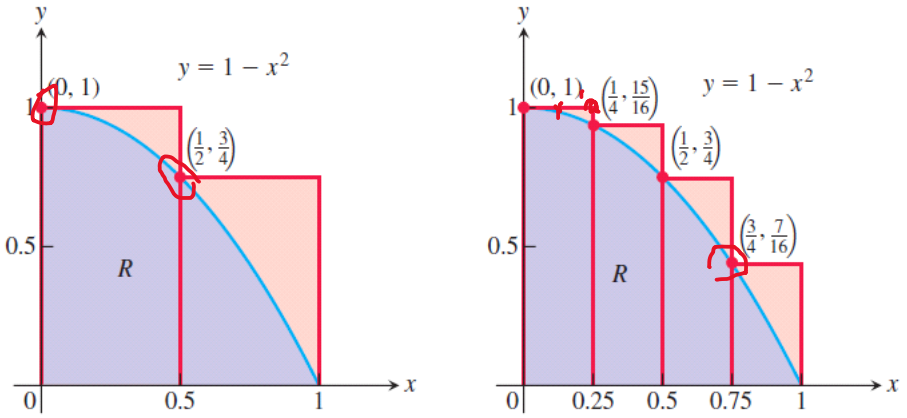
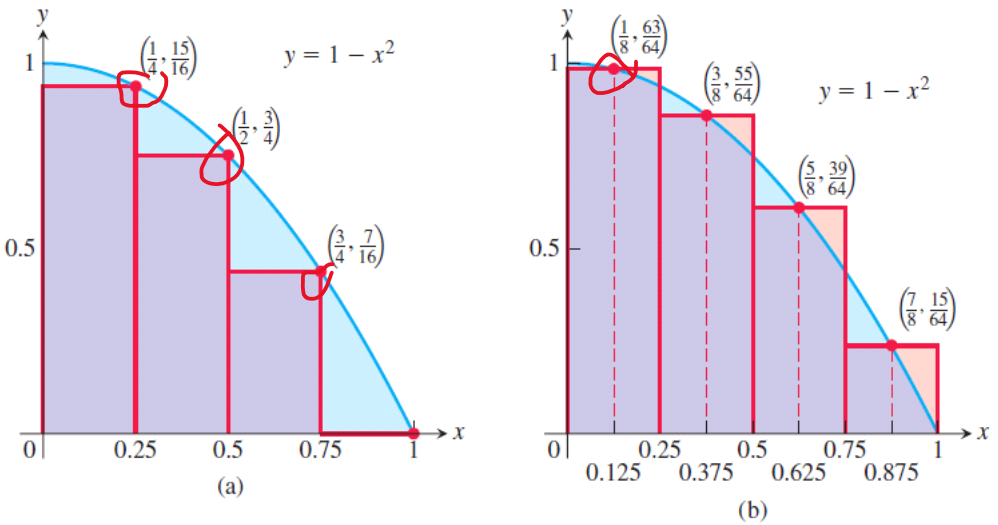


FIGURE 5.2 (a) We get an upper estimate of the area of R by using two rectangles containing R. (b) Four rectangles give a better upper estimate. Both estimates overshoot the true value for the area by the amount shaded in light red.

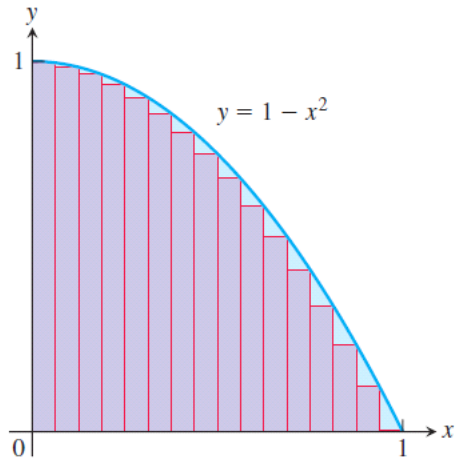
$$A \approx 1 \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{7}{8} = 0.875.$$

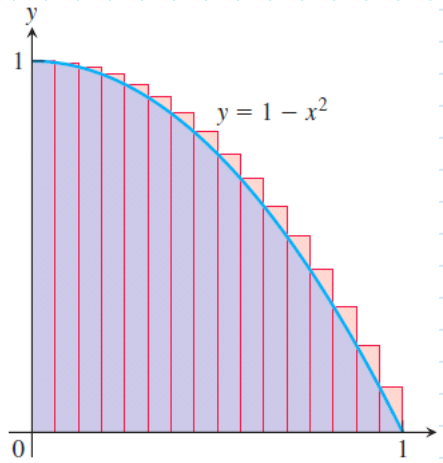
$$A \approx 1 \cdot \frac{1}{4} + \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4} = \frac{25}{32} = 0.78125,$$



$$A \approx \frac{15}{16}\cdot\frac{1}{4} + \frac{3}{4}\cdot\frac{1}{4} + \frac{7}{16}\cdot\frac{1}{4} + 0\cdot\frac{1}{4} = \frac{17}{32} = 0.53125.$$

$$0.53125 < A < 0.78125.$$





EXAMPLE 1 Table 5.1 shows the values of upper and lower sum approximations to the area of R, using up to 1000 rectangles. In Section 5.2 we will see how to get an exact value of the areas of regions such as R by taking a limit as the base width of each rectangle goes to zero and the number of rectangles goes to infinity. With the techniques developed there, we will be able to show that the area of R is exactly $2/3$. ■

TABLE 5.1 Finite approximations for the area of R

Number of subintervals	Lower sum	Midpoint sum	Upper sum
2	0.375	0.6875	0.875
4	0.53125	0.671875	0.78125
16	0.634765625	0.6669921875	0.697265625
50	0.6566	0.6667	0.6766
100	0.66165	0.666675	0.67165
1000	0.6661665	0.66666675	0.6671665

EXAMPLE 2 The velocity function of a projectile fired straight into the air is $f(t) = 160 - 9.8t$ m/sec. Use the summation technique just described to estimate how far the projectile rises during the first 3 sec. How close do the sums come to the exact value of 435.9 m? (You will learn how to compute the exact value easily in Section 5.4.)

TABLE 5.2 Travel-distance estimates

Number of subintervals	Length of each subinterval	Upper sum	Lower sum
3	1	450.6	421.2
6	1/2	443.25	428.55
12	1/4	439.58	432.23
24	1/8	437.74	434.06
48	1/16	436.82	434.98
96	1/32	436.36	435.44
192	1/64	436.13	435.67

EXAMPLE 4 Estimate the average value of the function $f(x) = \sin x$ on the interval $[0, \pi]$.

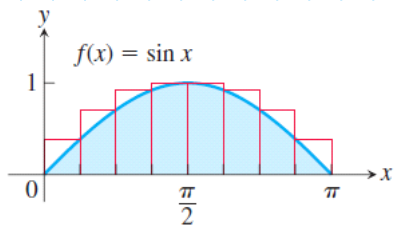


FIGURE 5.7 Approximating the area under $f(x) = \sin x$ between 0 and π to compute the average value of $\sin x$ over $[0, \pi]$, using eight rectangles (Example 4).

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$$A \approx \left(\sin\frac{\pi}{8} + \sin\frac{\pi}{4} + \sin\frac{3\pi}{8} + \sin\frac{\pi}{2} + \sin\frac{\pi}{2} + \sin\frac{5\pi}{8} + \sin\frac{3\pi}{4} + \sin\frac{7\pi}{8}\right) \cdot \frac{\pi}{8}$$

$$\approx (.38 + .71 + .92 + 1 + 1 + .92 + .71 + .38) \cdot \frac{\pi}{8} = (6.02) \cdot \frac{\pi}{8} \approx 2.364.$$

TABLE 5.5 Average value of $\sin x$ on $0 \le x \le \pi$

Number of subintervals	Upper sum estimate
8	0.75342
16	0.69707
32	0.65212
50	0.64657
100	0.64161
1000	0.63712

Riemann Sums

The theory of limits of finite approximations was made precise by the German mathematician Bernhard Riemann. We now introduce the notion of a *Riemann sum*, which underlies the theory of the definite integral studied in the next section.

We begin with an arbitrary bounded function f defined on a closed interval $[a, b]$. Like the function pictured in Figure 5.8, f may have negative as well as positive values. We subdivide the interval $[a, b]$ into subintervals, not necessarily of equal widths (or lengths), and form sums in the same way as for the finite approximations in Section 5.1. To do so, we choose $n - 1$ points $\{x_1, x_2, x_3, \ldots, x_{n-1}\}$ between a and b satisfying

$$a < x_1 < x_2 < \cdots < x_{n-1} < b.$$

To make the notation consistent, we denote a by x_0 and b by x_n, so that

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b.$$

The set

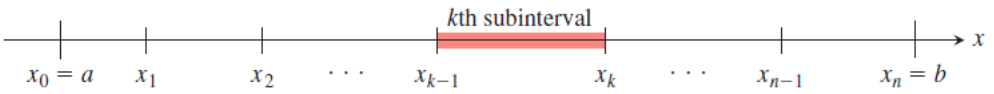
$$P = \{x_0, x_1, x_2, \ldots, x_{n-1}, x_n\}$$

is called a **partition** of $[a, b]$.

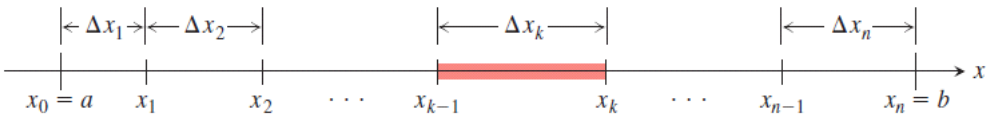
The partition P divides $[a, b]$ into n closed subintervals

$$[x_0, x_1], [x_1, x_2], \ldots, [x_{n-1}, x_n].$$

The first of these subintervals is $[x_0, x_1]$, the second is $[x_1, x_2]$, and the ***k*th subinterval** of P is $[x_{k-1}, x_k]$, for k an integer between 1 and n.

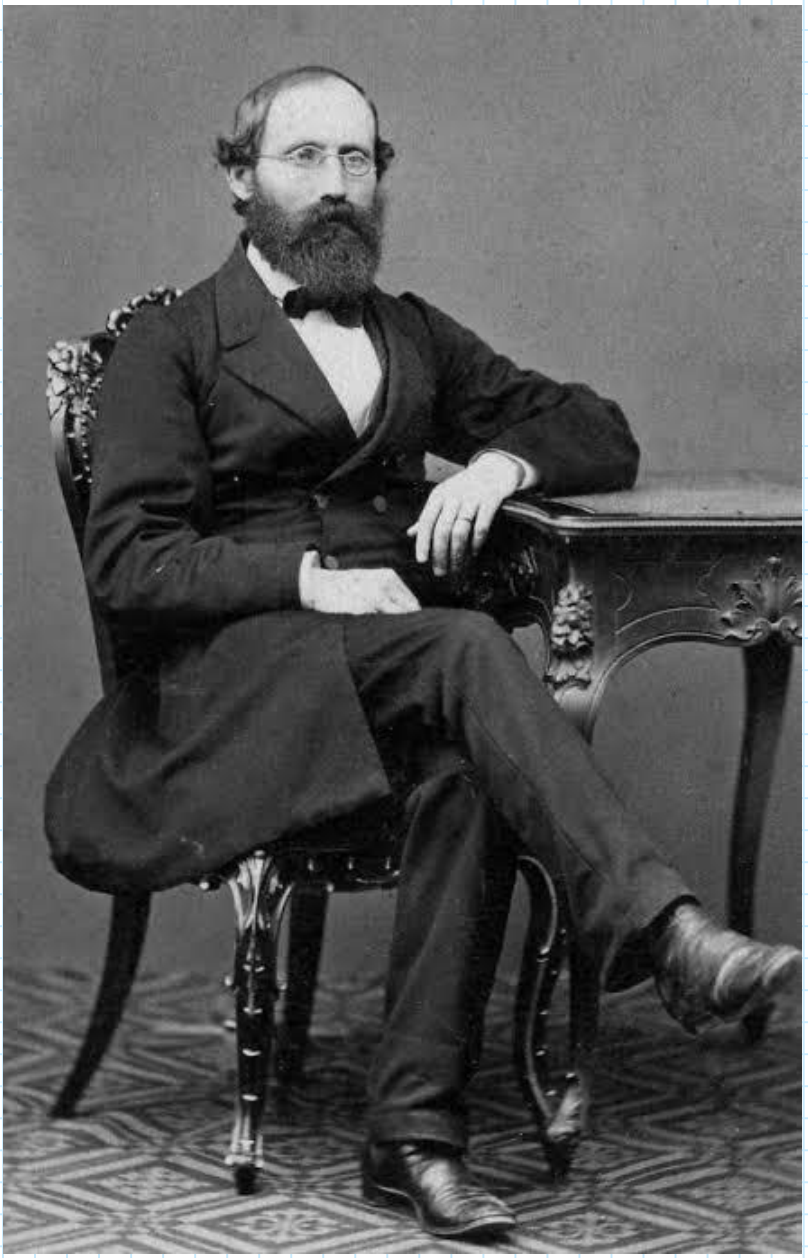


The width of the first subinterval $[x_0, x_1]$ is denoted Δx_1, the width of the second $[x_1, x_2]$ is denoted Δx_2, and the width of the kth subinterval is $\Delta x_k = x_k - x_{k-1}$. If all n subintervals have equal width, then the common width Δx is equal to $(b - a)/n$.

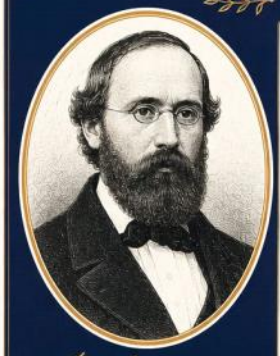


HISTORICAL BIOGRAPHY

Georg Friedrich Bernhard Riemann
(1826–1866)



INSPIRING MATHEMATICIANS



B. Riemann

BERNHARD RIEMANN

THE GEOMETER OF THE INFINITE

Riemann's profound insights connected geometry, analysis, and number theory in ways that reshaped our understanding of space, numbers, and the universe.

BORN
17 September 1826
Breselenz, Hanover, Germany

DIED
20 July 1866
Selasca, Italy (Age 39)

NATIONALITY
German

LEGACY
His ideas underpin much of modern mathematics, physics, and beyond.

ABOUT RIEMANN

Bernhard Riemann (1826–1866) was a German mathematician whose visionary ideas transformed modern mathematics and continue to influence science today.

He redefined geometry by exploring curved spaces, extended the concept of numbers to complex analysis, and laid deep foundations for number theory and topology.

Though he published only a few papers in his short life, the depth and beauty of his ideas opened new horizons for mathematics and physics.

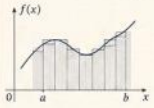
> *"Math is the science of order, and Riemann showed us that order can be curved, complex, and infinitely deep."*

KEY CONTRIBUTIONS

RIEMANNIAN GEOMETRY
Introduced the concept of curved space. His ideas now form the language of general relativity in physics.



RIEMANN INTEGRAL
Extended integration to more general functions, forming the foundation of modern real analysis.



RIEMANN ZETA FUNCTION
Defined the zeta function ζ(s), revealing deep patterns in prime numbers. His famous hypothesis remains one of mathematics' greatest unsolved problems.

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

COMPLEX ANALYSIS
Pioneered the study of complex functions on Riemann surfaces—a powerful tool in mathematics and engineering.



TOPOLOGY & MANIFOLDS
Laid foundations for topology and the study of manifolds, shaping modern geometry and many branches of mathematics.



SCIENTIFIC VISION
Riemann believed mathematics reveals the hidden structure behind nature, unifying diverse phenomena through deep ideas.

NOTABLE WORKS



- "On the Hypotheses Which Lie at the Bases of Geometry" (1854)
- "On the Number of Primes Less Than a Given Magnitude" (1859)
- "Über die Darstellbarkeit einer Funktion durch eine trigonometrische Reihe" (1867)

(Many works were published posthumously.)

BY THE NUMBERS

- ∞ Infinite impact from just a few papers.
- ζ Zeta function connects infinitely many primes.
- 39 A life of only 39 years, leaving an eternal legacy.

A LASTING LEGACY

From Einstein's theory of relativity to quantum physics, from number theory to data science—Riemann's ideas are the silent engine behind modern discovery.

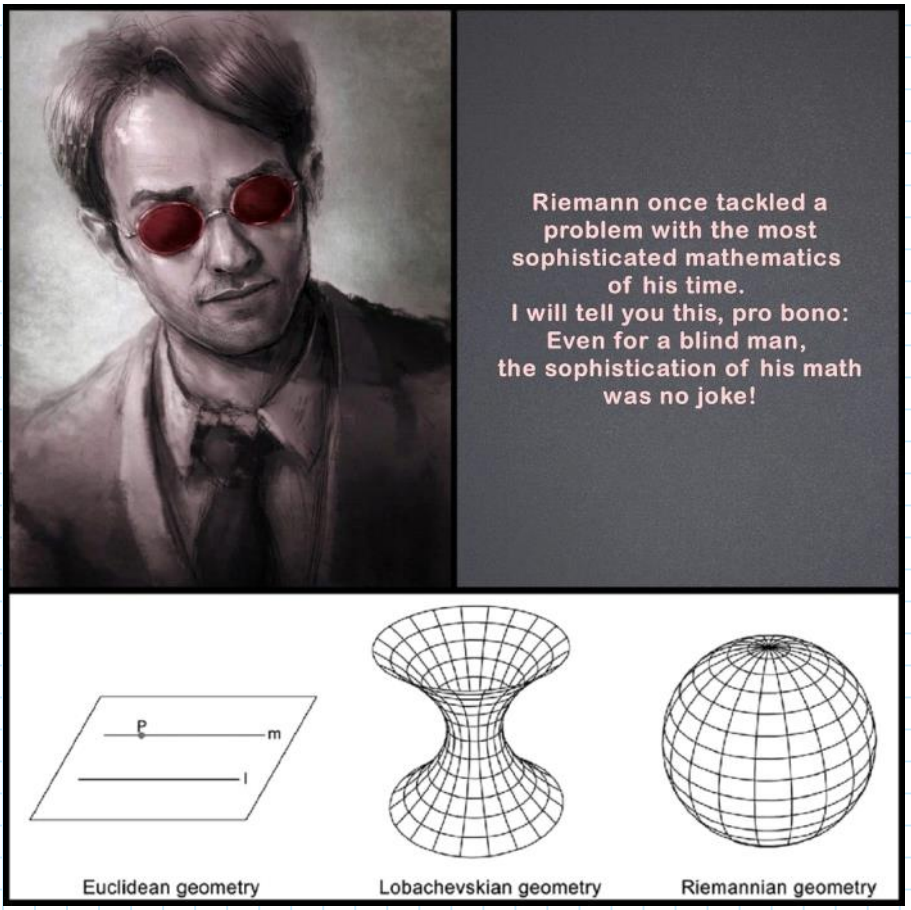


Riemann didn't just solve problems—he changed the questions we ask.

> *"The wonderful thing about mathematics is that its beauty is absolute."*
> — BERNHARD RIEMANN



CURATED BY DR. EMMANUEL JESUYON DANSU



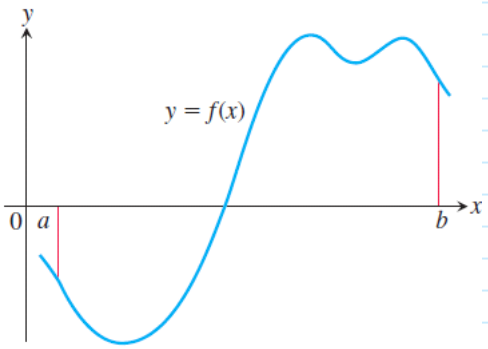
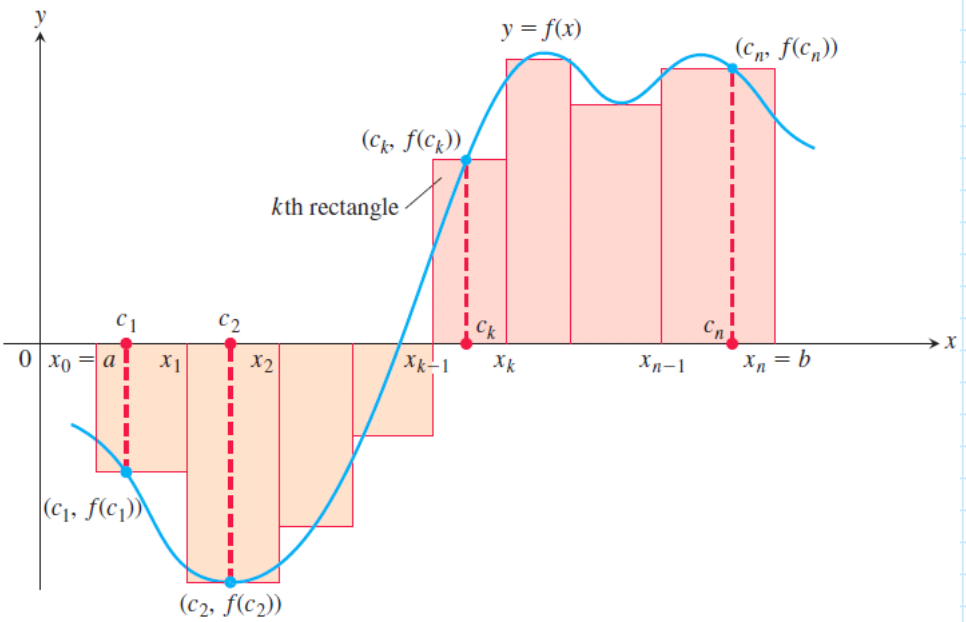


FIGURE 5.8 A typical continuous function $y = f(x)$ over a closed interval $[a, b]$.



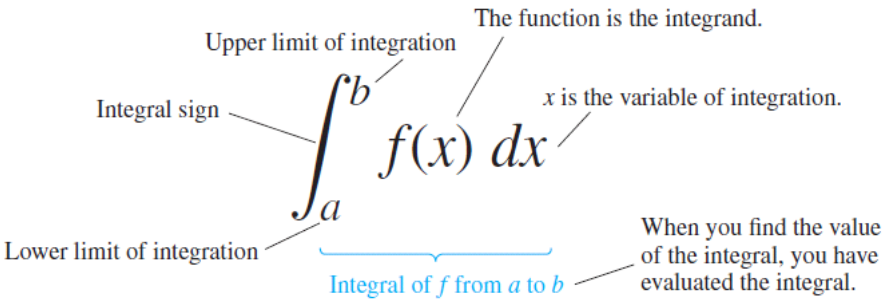
On each subinterval we form the product $f(c_k) \cdot \Delta x_k$. This product is positive, negative, or zero, depending on the sign of $f(c_k)$. When $f(c_k) > 0$, the product $f(c_k) \cdot \Delta x_k$ is the area of a rectangle with height $f(c_k)$ and width Δx_k. When $f(c_k) < 0$, the product $f(c_k) \cdot \Delta x_k$ is a negative number, the negative of the area of a rectangle of width Δx_k that drops from the x-axis to the negative number $f(c_k)$.

Finally we sum all these products to get

$$S_P = \sum_{k=1}^{n} f(c_k)\, \Delta x_k.$$

The sum S_P is called a **Riemann sum for f on the interval $[a, b]$**. There are many such

The Definite Integral



DEFINITION If $y = f(x)$ is nonnegative and integrable over a closed interval $[a, b]$, then the **area under the curve $y = f(x)$ over $[a, b]$** is the integral of f from a to b,

$$A = \int_a^b f(x)\, dx.$$

2 Definition of a Definite Integral If f is a function defined for $a \le x \le b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. We let $x_0 (= a), x_1, x_2, \ldots, x_n (= b)$ be the endpoints of these subintervals and we let $x_1^*, x_2^*, \ldots, x_n^*$ be any **sample points** in these subintervals, so x_i^* lies in the ith subinterval $[x_{i-1}, x_i]$. Then the **definite integral of f from a to b** is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points. If it does exist, we say that f is **integrable** on $[a, b]$.

1. *Order of Integration:* $\displaystyle\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$ — A definition

2. *Zero Width Interval:* $\displaystyle\int_a^a f(x)\,dx = 0$ — A definition when $f(a)$ exists

3. *Constant Multiple:* $\displaystyle\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$ — Any constant k

4. *Sum and Difference:* $\displaystyle\int_a^b (f(x) \pm g(x))\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$

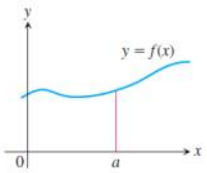
5. *Additivity:* $\displaystyle\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$

6. *Max-Min Inequality:* If f has maximum value max f and minimum value min f on $[a, b]$, then

$$\min f \cdot (b - a) \le \int_a^b f(x)\,dx \le \max f \cdot (b - a).$$

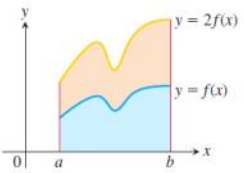
7. *Domination:* $f(x) \ge g(x)$ on $[a, b] \Rightarrow \displaystyle\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$

$f(x) \ge 0$ on $[a, b] \Rightarrow \displaystyle\int_a^b f(x)\,dx \ge 0$ (Special case)



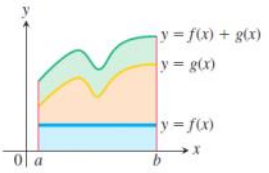
(a) *Zero Width Interval:*

$$\int_a^a f(x)\,dx = 0$$



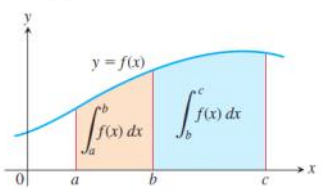
(b) *Constant Multiple:* $(k = 2)$

$$\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$$



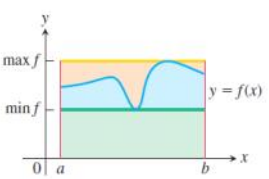
(c) *Sum: (areas add)*

$$\int_a^b (f(x) + g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$



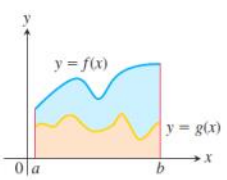
(d) *Additivity for Definite Integrals:*

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$$



(e) *Max-Min Inequality:*

$$\min f \cdot (b - a) \le \int_a^b f(x)\,dx \le \max f \cdot (b - a)$$



(f) *Domination:*

$f(x) \ge g(x)$ on $[a, b]$

$$\Rightarrow \int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$$

EXAMPLE 4 Compute $\int_0^b x\,dx$ and find the area A under $y = x$ over the interval $[0, b]$, $b > 0$.

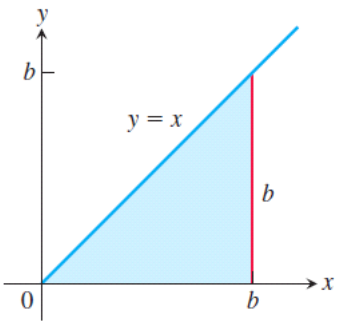


FIGURE 5.12 The region in Example 4 is a triangle.

DEFINITION If f is integrable on $[a, b]$, then its **average value on $[a, b]$**, also called its **mean**, is

$$\text{av}(f) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

EXAMPLE 5 Find the average value of $f(x) = \sqrt{4 - x^2}$ on $[-2, 2]$.

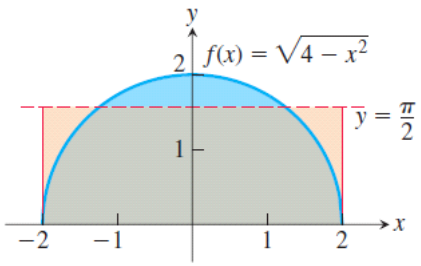


FIGURE 5.15 The average value of $f(x) = \sqrt{4 - x^2}$ on $[-2, 2]$ is $\pi/2$ (Example 5). The area of the rectangle shown here is $4 \cdot (\pi/2) = 2\pi$, which is also the area of the semicircle.

Solution We recognize $f(x) = \sqrt{4 - x^2}$ as a function whose graph is the upper semicircle of radius 2 centered at the origin (Figure 5.15).

Since we know the area inside a circle, we do not need to take the limit of Riemann sums. The area between the semicircle and the x-axis from -2 to 2 can be computed using the geometry formula

$$\text{Area} = \frac{1}{2} \cdot \pi r^2 = \frac{1}{2} \cdot \pi(2)^2 = 2\pi.$$

Because f is nonnegative, the area is also the value of the integral of f from -2 to 2,

$$\int_{-2}^{2} \sqrt{4 - x^2}\, dx = 2\pi.$$

Therefore, the average value of f is

$$\text{av}(f) = \frac{1}{2 - (-2)} \int_{-2}^{2} \sqrt{4 - x^2}\, dx = \frac{1}{4}(2\pi) = \frac{\pi}{2}.$$

Notice that the average value of f over $[-2, 2]$ is the same as the height of a rectangle over $[-2, 2]$ whose area equals the area of the upper semicircle (see Figure 5.15). ■

The Fundamental Theorem of Calculus

HISTORICAL BIOGRAPHY
Sir Isaac Newton
(1642–1727)

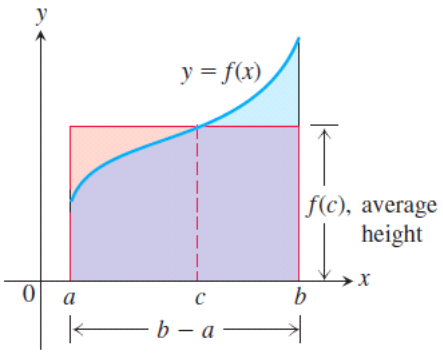


FIGURE 5.16 The value $f(c)$ in the Mean Value Theorem is, in a sense, the average (or *mean*) height of f on $[a, b]$. When $f \geq 0$, the area of the rectangle is the area under the graph of f from a to b,

$$f(c)(b - a) = \int_a^b f(x)\, dx.$$

THEOREM 3—The Mean Value Theorem for Definite Integrals If f is continuous on $[a, b]$, then at some point c in $[a, b]$,

$$f(c) = \frac{1}{b - a} \int_a^b f(x)\, dx.$$

THEOREM 4—The Fundamental Theorem of Calculus, Part 1 If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t)\,dt$ is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$:

$$F'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x). \tag{2}$$

EXAMPLE 2 Use the Fundamental Theorem to find dy/dx if

(a) $y = \int_a^x (t^3 + 1)\,dt$ **(b)** $y = \int_x^5 3t \sin t\,dt$

(c) $y = \int_1^{x^2} \cos t\,dt$ **(d)** $y = \int_{1+3x^2}^4 \frac{1}{2 + e^t}\,dt$

$$y = \int_1^u \cos t\,dt \quad \text{and} \quad u = x^2.$$

We must therefore apply the Chain Rule when finding dy/dx.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du}\cdot\frac{du}{dx}\\ &= \left(\frac{d}{du}\int_1^u \cos t\,dt\right)\cdot\frac{du}{dx}\\ &= \cos u\cdot\frac{du}{dx}\\ &= \cos(x^2)\cdot 2x\\ &= 2x\cos x^2\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}\int_{1+3x^2}^4 \frac{1}{2 + e^t}\,dt &= \frac{d}{dx}\left(-\int_4^{1+3x^2} \frac{1}{2 + e^t}\,dt\right)\\ &= -\frac{d}{dx}\int_4^{1+3x^2} \frac{1}{2 + e^t}\,dt\\ &= -\frac{1}{2 + e^{(1+3x^2)}}\frac{d}{dx}(1 + 3x^2)\\ &= -\frac{6x}{2 + e^{(1+3x^2)}}\end{aligned}$$

THEOREM 4 (Continued)—The Fundamental Theorem of Calculus, Part 2 If f is continuous over $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

(a) $\int_0^\pi \cos x\,dx$

(b) $\displaystyle\int_{-\pi/4}^{0} \sec x \tan x \, dx$

(c) $\displaystyle\int_{1}^{4} \left(\frac{3}{2}\sqrt{x} - \frac{4}{x^2}\right) dx =$

(d) $\displaystyle\int_{0}^{1} \frac{dx}{x+1} =$

$=$

(e) $\displaystyle\int_{0}^{1} \frac{dx}{x^2+1} =$

$=$

THEOREM 5—The Net Change Theorem The net change in a differentiable function $F(x)$ over an interval $a \le x \le b$ is the integral of its rate of change:

$$F(b) - F(a) = \int_a^b F'(x)\,dx. \tag{6}$$

EXAMPLE 5 Consider again our analysis of a heavy rock blown straight up from the ground by a dynamite blast (Example 3, Section 5.1). The velocity of the rock at any time t during its motion was given as $v(t) = 160 - 32t$ ft/sec.

(a) Find the displacement of the rock during the time period $0 \le t \le 8$.

(b) Find the total distance traveled during this time period.